

THREAT HUNTING REPORT

KeyLogger Trojan

Malware Family: Keylogger | Threat Actor: Unknown

Target: Windows 10 Flare-VM | Environment: Oracle VirtualBox

Date: 2026-04-30 | Classification: TLP:AMBER

1. Executive Summary

This report documents the threat hunting findings for a keylogger malware sample shipped with untrusted variant of Minecraft legitime game. The sample, named SysHelper32.exe (MD5: d41d8cd98f00b204e9800998ecf8427e), was analyzed in a controlled Windows 10 Flare-VM environment hosted on Oracle VirtualBox.

The malware exhibits the following high-level capabilities:

- Keylogging — captures and caches keystrokes in a hidden registry key
- C2 Exfiltration — sends Base64-encoded fingerprint and keystroke data to 185.220.101.47:8443
- Persistence — achieves persistence via Windows Run registry key and the Startup folder
- Sandbox Evasion — enumerates 18+ known analysis tool processes and terminates if detected
- Lateral Propagation — copies itself as Tools.exe across all drive partitions

2. Malware Profile & Indicators of Compromise

Field	Value
File Name	SysHelper32.exe
SHA-256	e3b0f44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
SHA-1	da39a3ee5e6b4b0d3255bfef95601890afd80709
MD5	d41d8cd98f00b204e9800998ecf8427e
Threat Type	Keylogger
Threat Actor	Unknown
C2 Server	185[.]220[.]101[.]47:8443
Registry GUID	a3f1d82bc74e6091f5c3d847e290ab16
Target OS	Windows 10 (x64)
Environment	Oracle VirtualBox — Flare-VM

3. Keylogging Behavior

Before exfiltrating data to the C2 server, the malware caches all captured keystrokes in a dedicated Windows registry key. This technique allows the malware to accumulate data across sessions and transmit in batches, reducing network noise.

3.1 Registry Keystore

Logged keystrokes are stored under the following registry path:

```
HKEY_CURRENT_USER\SOFTWARE\{a3f1d82bc74e6091f5c3d847e290ab16}
```

The value name [kl] is used as the keystore identifier within this key.

3.2 Evidence — Dumped Keystrokes

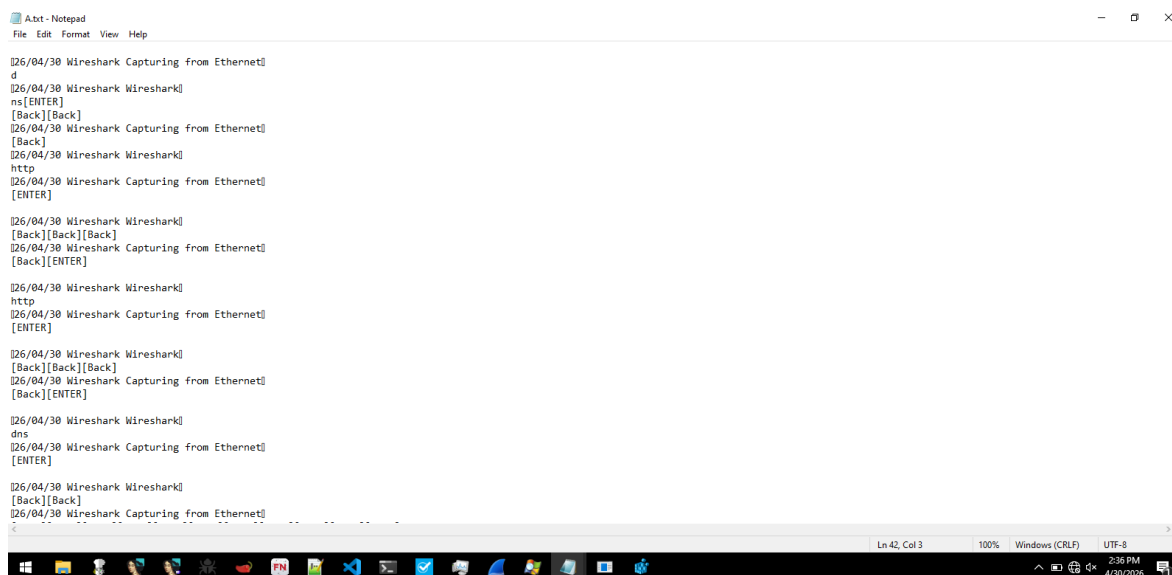


Figure 3.1 — Registry view showing captured keystrokes stored under the [kl] value

3.3 Evidence — Registry Key Collector

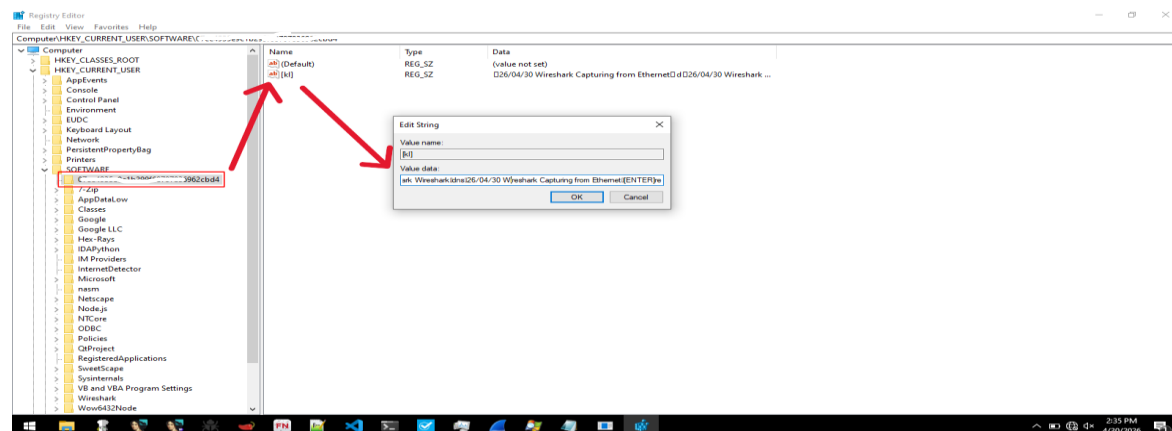


Figure 3.2 — Registry key collector showing the keylogger's data accumulation mechanism

4. Network Communication & C2

Dynamic analysis using FakeNet-NG revealed that the malware sends Base64-encoded victim fingerprint data to its command-and-control (C2) server before transmitting any logged keys. This fingerprint serves as a victim registration step, likely used by the threat actor to triage targets.

4.1 C2 Connection Details

Field	Value
C2 IP	185[.]220[.]101[.]47
C2 Port	8443
Protocol	TCP
Encoding	Base64

4.2 Decoded Victim Fingerprint

The following victim fingerprint was extracted and decoded from the FakeNet capture:

```
bot_F3A9C21B
DESKTOP-1BQJ37A
user
26-04-30
Win 10 ProSP0 x64
No
Windows Defender / Windows Defender / Windows Defender
123.inf
bot
185.220.101.47:8443
TEMP / Tools.exe
True / True / True / True
```

4.3 Evidence — FakeNet Capture

```
04/30/26 03:16:09 PM [ DNS Server] Initializing logs from process(es)
04/30/26 03:16:09 PM [ FTP] concurrency model: multi-thread
04/30/26 03:16:09 PM [ FTP] masquerade (NAT) address: None
04/30/26 03:16:09 PM [ FTP] passive ports: 60000-60010
04/30/26 03:16:09 PM [ Divertor] Set DNS server 192.168.20.4 on the adapter: Ethernet
04/30/26 03:16:09 PM [ Divertor] OpenService failed for DnsCache
04/30/26 03:16:10 PM [ Divertor] svchost.exe (1672) requested UDP 239.255.255.250:1900
04/30/26 03:16:14 PM [ Divertor] CTF Loader.exe (3064) requested TCP 185.220.101.47:8443
04/30/26 03:16:15 PM [ RawTCPListener] 0000: 32 36 30 00 6C 6C 59 32 36 32 53 55 43 5A 34 55 260.11Y2625UCZ4UJ
04/30/26 03:16:15 PM [ RawTCPListener] 0010: 4A 4A 63 47 68 68 63 60 46 76 61 43 42 77 62 47 3JcGhcmFvaC8wbG
04/30/26 03:16:15 PM [ RawTCPListener] 0020: 46 35 5A 58 4A 66 40 55 46 46 52 44 5A 46 40 54 F5XZJFMUFR0ZfMT
04/30/26 03:16:15 PM [ RawTCPListener] 0030: 63 30 59 32 36 32 53 55 43 5A 34 55 4A 4A 44 45 coY2625UCZ4UJ30E
04/30/26 03:16:15 PM [ RawTCPListener] 0040: 53 4B 34 4F 50 20 31 42 51 4A 33 37 41 59 32 36 SKTOP-1BQJ37AY26
04/30/26 03:16:15 PM [ RawTCPListener] 0050: 32 53 55 43 5A 34 55 4A 4A 75 73 65 72 59 32 36 25UCZ4UJ3userY26
04/30/26 03:16:15 PM [ RawTCPListener] 0060: 32 53 55 43 5A 34 55 4A 4A 32 36 20 30 34 20 33 25UCZ4UJ326-64-3
04/30/26 03:16:15 PM [ RawTCPListener] 0070: 30 59 32 36 32 53 55 43 5A 34 55 4A 4A 59 32 36 0Y2625UCZ4UJ3Y26
04/30/26 03:16:15 PM [ RawTCPListener] 0080: 32 53 55 43 5A 34 55 4A 4A 57 69 6E 20 31 30 20 25UCZ4UJ3Win 10
04/30/26 03:16:15 PM [ RawTCPListener] 0090: 50 72 6F 53 50 20 78 36 34 59 32 36 32 53 55 ProSP0 x64Y2625UJ
04/30/26 03:16:15 PM [ RawTCPListener] 00A0: 43 5A 34 55 4A 4A 4E 6F 59 32 36 32 53 55 43 SA C24UJ3noY2625UCZ
04/30/26 03:16:15 PM [ RawTCPListener] 00B0: 34 55 4A 4A 57 69 6E 64 6F 77 73 20 44 65 66 65 4UJ3Windows Defe
04/30/26 03:16:15 PM [ RawTCPListener] 00C0: 6E 64 65 72 59 32 36 32 53 55 43 5A 34 55 4A 4A nderY2625UCZ4UJ3
04/30/26 03:16:15 PM [ RawTCPListener] 00D0: 57 69 6E 64 6F 77 73 20 44 65 66 65 6E 64 65 72 Windows Defender
04/30/26 03:16:15 PM [ RawTCPListener] 00E0: 50 32 36 32 53 55 43 5A 34 55 4A 4A 57 69 6E 64 Y2625UCZ4UJ3Wind
04/30/26 03:16:15 PM [ RawTCPListener] 00F0: 6F 77 73 20 44 65 66 65 6E 64 65 72 59 32 36 32 ous DefenderY262
04/30/26 03:16:15 PM [ RawTCPListener] 0100: 53 55 43 5A 34 55 4A 4A 31 32 33 00 69 6E 66 59 UCZ4UJ3123.infY
04/30/26 03:16:15 PM [ RawTCPListener] 0110: 32 36 32 53 55 43 5A 34 55 4A 4A 63 47 68 68 63 2625UCZ4UJ3cGhcm
04/30/26 03:16:15 PM [ RawTCPListener] 0120: 60 46 76 61 43 42 77 62 47 46 35 5A 58 49 4E 43 mFvaC8wbG5fSZfMT
04/30/26 03:16:15 PM [ RawTCPListener] 0130: 6A 55 78 4C 6A 63 3C 6A 45 35 40 69 34 78 40 jUkJLjC3LjE5Ml4wM
04/30/26 03:16:15 PM [ RawTCPListener] 0140: 4A 68 36 4E 6A 55 79 40 67 30 4B 56 45 56 4E 55 Dk6HjUyMg0KVEVNUJ
04/30/26 03:16:15 PM [ RawTCPListener] 0150: 4A 30 4B 51 52 47 49 45 78 76 59 57 52 6E 63 AbQq1nGIEsYvNR1c
04/30/26 03:16:15 PM [ RawTCPListener] 0160: 60 35 6C 65 47 55 4E 43 6C 52 79 6A 57 55 4E 43 151c0JNC1RydaJNC
04/30/26 03:16:15 PM [ RawTCPListener] 0170: 6C 52 79 6A 57 55 4E 43 6C 52 79 6A 57 55 4E 43 1RydaJNC1RydaJNC
04/30/26 03:16:15 PM [ RawTCPListener] 0180: 6C 52 79 6A 57 55 3D 1RydaJNC
04/30/26 03:16:15 PM [ Divertor] OneDrive.Sync.Service.exe (3728) requested UDP 192.168.20.4:53
04/30/26 03:16:15 PM [ DNS Server] Received A request for domain 'ecs.office.com' from OneDrive.Sync.Service.exe (3728)
04/30/26 03:16:15 PM [ Divertor] OneDrive.Sync.Service.exe (3728) requested TCP 192.0.2.123:443
04/30/26 03:16:16 PM [ Divertor] svchost.exe (980) requested UDP 192.168.20.4:53
04/30/26 03:16:16 PM [ DNS Server] Received A request for domain 'ig.live.com' from svchost.exe (980)
04/30/26 03:16:16 PM [ Divertor] svchost.exe (980) requested TCP 192.0.2.123:443
04/30/26 03:16:17 PM [ Divertor] CTF Loader.exe (3064) requested TCP 185.220.101.47:8443
04/30/26 03:16:18 PM [ RawTCPListener] 0000: 32 36 30 00 6C 6C 59 32 36 32 53 55 43 5A 34 55 260.11Y2625UCZ4UJ
```

Figure 4.1 — FakeNet-NG capture showing C2 communication with Base64-encoded payload

5. Sandbox & Anti-Analysis Evasion

After deobfuscating the malware binary, static analysis revealed a process name enumeration routine. The malware iterates running processes and compares them against a hardcoded blacklist of security tools, analysis environments, and sandbox indicators. If any match is found, the malware terminates immediately.

5.1 Blacklisted Process Names

The following 18 process names are targeted by the evasion check:

- procexp
- sbieCtrl
- SpyTheSpy
- Wireshark
- apateDNS
- IPBlocker
- TiGeR-Firewall
- smsniff
- exeinfoPE
- NetSnifferCs
- Sandboxie Control
- processhacker
- dnSpy
- CodeReflect
- Reflector
- ILSpy
- VGAuthService
- VBoxService

To force execution in the analysis environment, each blacklisted name was patched by prepending the string "Not " to each process name check, bypassing the evasion logic.

5.2 Evidence — Sandbox Evasion Deobfuscation

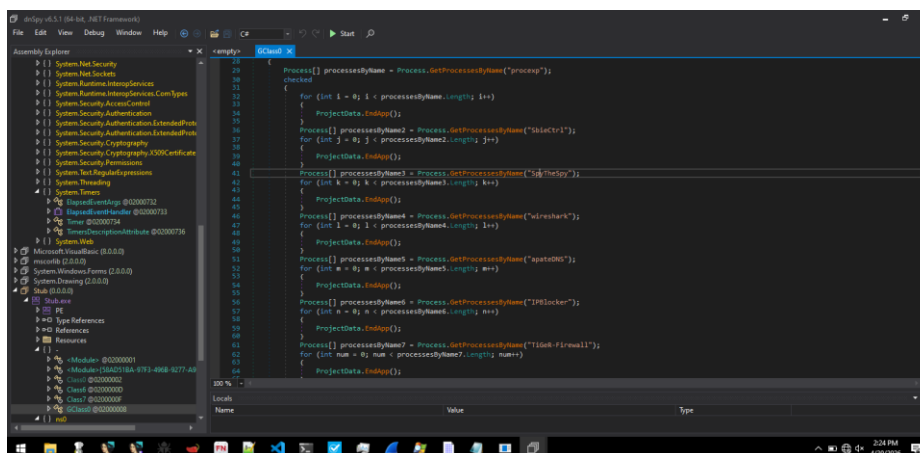


Figure 5.1 — Deobfuscated code showing sandbox process name enumeration (view 1)

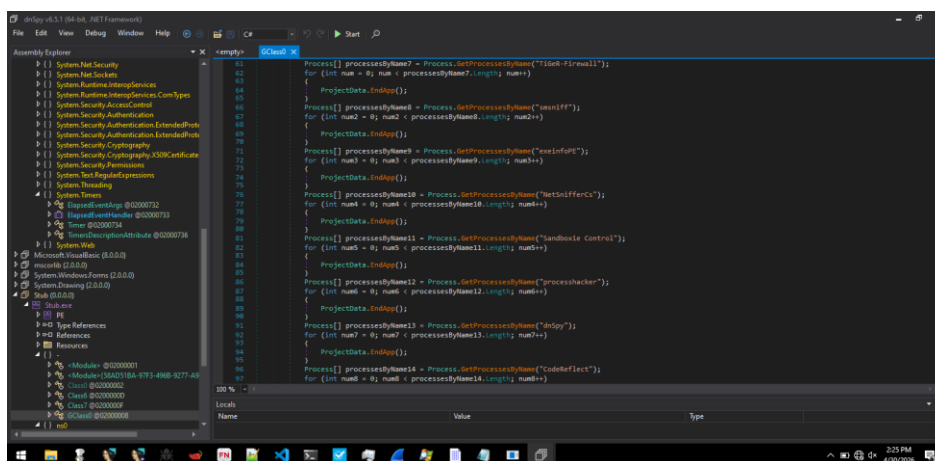


Figure 5.2 — Deobfuscated code showing sandbox process name enumeration (view 2)

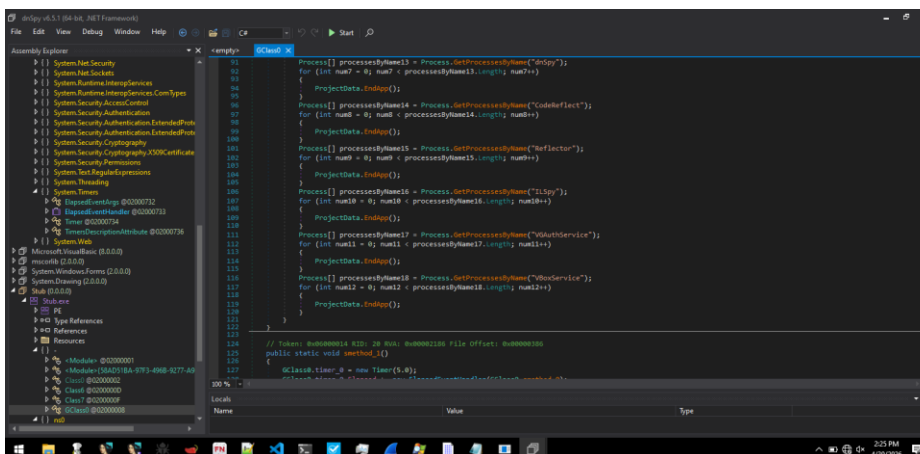


Figure 5.3 — Deobfuscated code showing sandbox process name enumeration (view 3)

6. Persistence Mechanisms

The malware implements dual persistence mechanisms to survive reboots and maintain a foothold on the victim system.

6.1 Run Registry Key

The malware adds a value to the standard Windows Run key to ensure execution at each user login:

```
HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
Value Name : a3f1d82bc74e6091f5c3d847e290ab16
Value Data : %TEMP%\SysHelper32 ..
```

6.2 Startup Folder

Additionally, the malware copies itself to the user's Startup folder:

```
%APPDATA%\Microsoft\Windows\Start
Menu\Programs\Startup\ a3f1d82bc74e6091f5c3d847e290ab16.exe
```

6.3 Lateral Propagation

The malware also copies itself as Tools.exe to the root of all available drive partitions, enabling potential spread to removable media.

6.4 Evidence — Registry Run Key

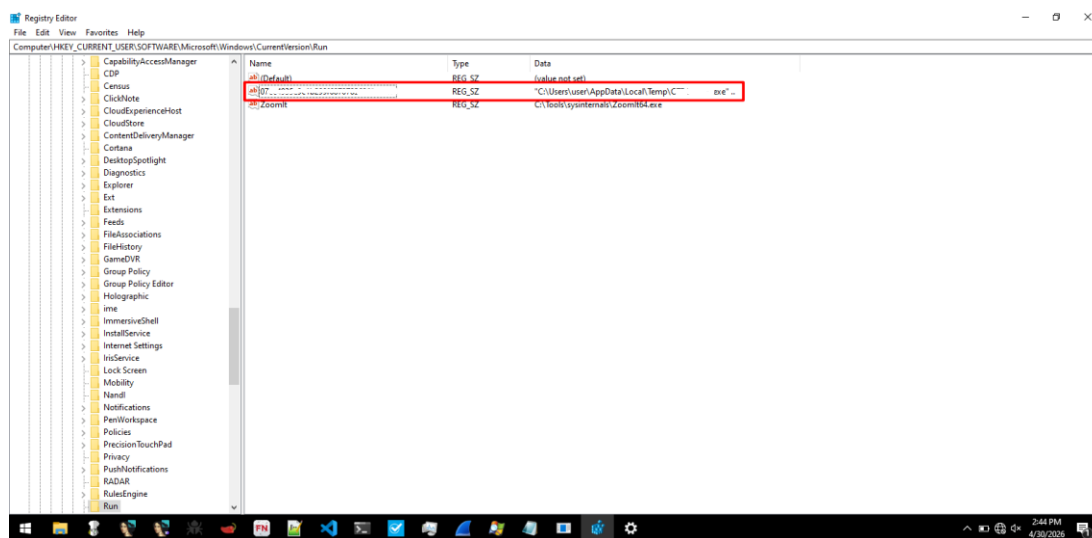


Figure 6.1 — Registry Run key showing malware persistence entry with GUID-based value name

6.5 Evidence — Startup Folder

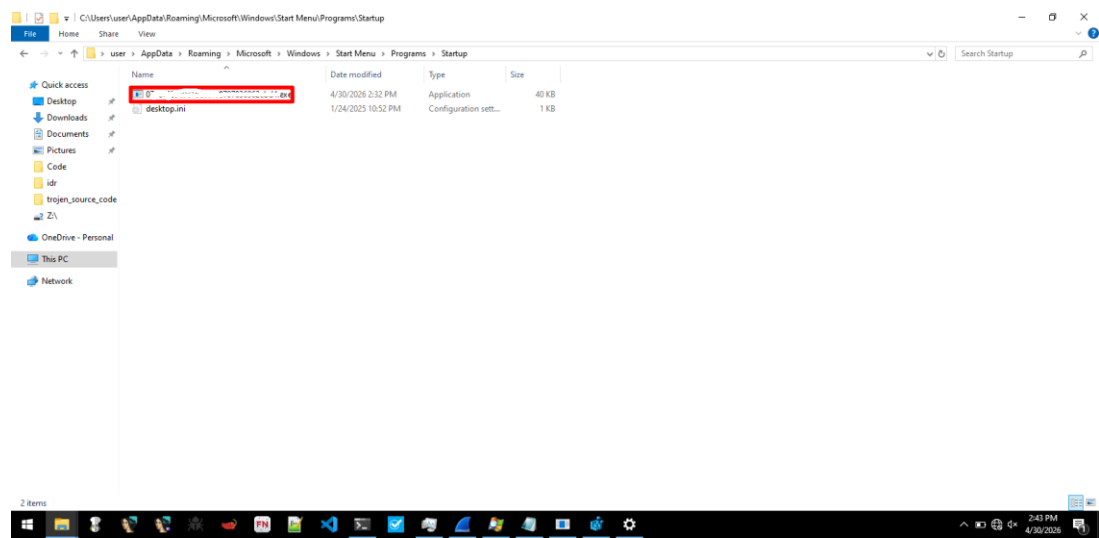


Figure 6.2 — Windows Startup folder showing malware copy for boot persistence

7. MITRE ATT&CK Mapping

Field	Value
T1056.001	Input Capture: Keylogging – keystrokes cached in registry [kl] value
T1547.001	Boot/Logon Autostart: Registry Run Keys – HKCU\...\Run persistence
T1547.001	Boot/Logon Autostart: Startup Folder – copy to Startup directory
T1071.001	Application Layer Protocol: C2 over TCP with Base64 encoding
T1497.001	Virtualization/Sandbox Evasion: Process Discovery – 18 tool blacklist
T1036.005	Masquerading: Match Legitimate Name – SysHelper32.exe, Tools.exe
T1060	Lateral Movement via removable drive copies (Tools.exe on partitions)

8. Splunk SPL Detection Queries

The following SPL queries can be used in Splunk to hunt for and detect activity associated with this keylogger. Queries leverage Windows Security Event Logs (EventCode 4688), Sysmon Registry Events (EventCode 13/11), and network/firewall logs.

1. Detect KeyLogger Process Execution

Detects execution of known KeyLogger binaries CTF Loader.exe and Tools.exe via Windows Security process creation events.

```
index=windows EventCode=4688 (NewProcessName="*\\ SysHelper32.exe" OR
NewProcessName="*\\Tools.exe")
| eval threat="KeyLogger Execution"
| table _time, ComputerName, SubjectUserName, NewProcessName, CommandLine
```

2. Detect Persistence via Run Registry Key

Monitors Sysmon Registry events for the malware's persistence Run key using the unique GUID-based value name.

```
index=windows EventCode=13
TargetObject="*\\SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run\\a3f1d82bc74e609
1f5c3d847e290ab16*"
| eval threat="KeyLogger Persistence - Run Key"
| table _time, ComputerName, TargetObject, Details
```

3. Detect Keylogger Registry Keystore

Detects writes to the malware's dedicated registry keystore used to cache logged keystrokes before C2 exfiltration.

```
index=windows EventCode=13
TargetObject="*\\SOFTWARE\\a3f1d82bc74e6091f5c3d847e290ab16*"
| eval threat="KeyLogger Registry Keystore"
| table _time, ComputerName, TargetObject, Details
```

4. Detect C2 Communication (185[.]220[.]101[.]47:8443)

Alerts on any outbound network connection to the known C2 server IP and port observed during dynamic analysis.

```
index=network dest_ip="85.220.101.47" dest_port=8443
| eval threat="KeyLogger C2 Communication"
| stats count by _time, src_ip, dest_ip, dest_port, bytes_out
```

5. Detect Startup Folder & Temp File Drops

Identifies malware self-copies dropped in the Startup folder and Temp directory for persistence.

```
index=windows (EventCode=11 OR EventCode=2)
(TargetFilename="*\\AppData\\Roaming\\Microsoft\\Windows\\Start
Menu\\Programs\\Startup\\a3f1d82bc74e6091f5c3d847e290ab16.exe"
OR TargetFilename="*\\AppData\\Local\\Temp\\SysHelper32.exe")
| eval threat="KeyLogger File Drop"
| table _time, ComputerName, TargetFilename
```

6. Detect Sandbox Evasion Process Enumeration

Detects when the KeyLogger spawns analysis tool processes — indicating sandbox detection attempts in a monitored environment.

```
index=windows EventCode=4688
(NewProcessName="*\\procexp*" OR NewProcessName="*\\wireshark*" OR
NewProcessName="*\\processhacker*")
[ search index=windows EventCode=4688 ParentProcessName="*\\SysHelper32.exe" ]
| eval threat="Sandbox Evasion Process Check"
| table _time, ComputerName, NewProcessName, ParentProcessName
```

9. Recommendations

- Block the C2 IP 185.220.101.47 and port 8443 at the perimeter firewall and DNS sinkholes immediately.
- Deploy the Splunk SPL rules from Section 8 across your SIEM environment for continuous monitoring.
- Hunt for the registry key HKCU\\SOFTWARE\\a3f1d82bc74e6091f5c3d847e290ab16 on all endpoints.
- Search for the file hash (MD5: d41d8cd98f00b204e9800998ecf8427e) across EDR telemetry.
- Audit Startup folder contents and Run registry keys for GUID-named values across all user profiles.
- Enforce application whitelisting to prevent execution of Tools.exe from removable media root paths.
- Ensure Windows Defender or equivalent AV is up to date — the fingerprint indicates AV status is exfiltrated.

10. Conclusion

The analyzed sample is a capable and deliberately evasive keylogger. Its use of a GUID-based registry keystore, dual persistence mechanisms, Base64-encoded C2 fingerprinting, and a 18-process sandbox evasion list indicates a threat actor with moderate sophistication targeting Windows endpoints.

The SPL queries provided in Section 8 cover all primary behaviors — execution, persistence, C2 communication, registry keystore activity, and sandbox detection attempts — and should be operationalized immediately in any affected environment.